

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	Abdul Wahid. H	Reg. No.:	192321079
Section / Batch:	B.Tech IT	Date:	30/07/2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A smart factory stores the hourly production count of eight machines in consecutive memory locations. The supervisor wants to identify the machine with the highest production and determine whether the total production exceeds the daily target stored in another memory location, Design and implement an 8085 ALP to:
 - Calculate the total production.
 - Find the highest production value.
 - Compare the total with the target value.
 - Store a status flag indicating whether the target has been achieved.
2. An automated chemical plant records ten temperature readings. A reading is considered abnormal if it is greater than the predefined threshold stored in memory. The controller must count the number of abnormal readings and identify the highest recorded temperature. Write an 8085 ALP that
 - Scans all sensor readings.
 - Counts the number of abnormal readings.
 - Finds the maximum temperature.
 - Stores both results in memory.
3. An educational institution stores marks of ten students in memory. A student is considered successful if the mark is greater than or equal to the pass mark stored in memory. Develop an 8085 ALP to
 - Count the number of students who passed.
 - Find the highest mark.
 - Calculate the total marks obtained by all students.
 - Store all results in separate memory locations.

Code of Conduct

I Abdul Wahid. H / 192321079 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
/ 4	/ 4	/ 4	/ 4	
/ 4	/ 4	/ 4	/ 4	
/ 4	/ 4	/ 4	/ 4	
/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

1. A smart factory stores the hourly production count of eight machines in consecutive memory locations. The supervisor wants to identify the machine with the highest production and determine whether the total production exceeds the daily target stored in another memory location, Design and implement an 8085 ALP to:

- Calculate the total production.
- Find the highest production value.
- Compare the total with the target value.
- Store a status flag indicating whether the target has been achieved.

Load me at 2000

```

1  LXI H, 8500H
2  MVI B, 08H
3  MVI D, 00H
4  MVI E, 00H
5  MOV C, 00H
6
7  LOOP: MOV A, M
8        MOV B, A
9        ADD E
10       MOV E, A
11       JNC NOCARRY
12       INR D
13  NOCARRY:
14       MOV A, B
15       CMP C
16       JC SKIP
17       MOV C, A
18  SKIP:
19       INX H
20       DCR B
21       JNZ LOOP
22
23       MOV A, E
24       STA 8509H
25       MOV A, D
26       STA 850AH
27       MOV A, C
28       STA 850BH
29       LDA 8508H
30       MOV B, A
31       MOV A, D
32       CPI 00H
33       JNZ ACHIEVED
34       MOV A, E
35       CMP B
36       JC NOTACHIEVED
37  ACHIEVED:
38       MVI A, 01H
39       STA 850CH
40       JMP END
41  NOTACHIEVED:
42       MVI A, 00H
43       STA 850CH
44  END:
45       HLT

```

Data Stack KeyPad Memory

Start 8500

Address (Hex)	Address	Data
2134	8500	35
2135	8501	42
2136	8502	28
2137	8503	4B
2138	8504	30
2139	8505	3C
213A	8506	27
213B	8507	45
213C	8508	12C
213D	8509	00
213F	850B	00
2140	850C	00

Line No Assembler Message

0 Program assembled successfully

Memory Locations :
8500H – 8507H : Production count of 8 machines
8508H : Daily target
8509H – 850AH : Total production (Low, High)
850BH : Highest production value
850CH : Status flag (01H- Achieved, 00H- Not Achieved)

2. An automated chemical plant records ten temperature readings. A reading is considered abnormal if it is greater than the predefined threshold stored in memory. The controller must count the number of abnormal readings and identify the highest recorded temperature. Write an 8085 ALP that

- Scans all sensor readings.
- Counts the number of abnormal readings.
- Finds the maximum temperature.
- Stores both results in memory.

Load me at 2000

```

1  LXI H,8500H
2  MVI E,0AH
3  MVI D,00H
4  MOV A,M
5  MOV C,A
6
7  LOOP: MOV A,M
8        MOV B,A
9        LDA 850AH
10       CMP B
11       JC ABNORMAL
12       JZ CHECKMAX
13       JMP CHECKMAX
14
15  ABNORMAL: INR D
16
17  CHECKMAX: MOV A,B
18            CMP C
19            JC NEXT
20            MOV C,A
21
22  NEXT:
23        INX H
24        DCR E
25        JNZ LOOP
26
27  MOV A,D
28  STA 850BH
29  MOV A,C
30  STA 850CH
31  HLT

```

Data
Stack
Keypad
Memory

Start 8500

Address (Hex)	Address	Data
2134	8500	62
2135	8501	70
2136	8502	68
2137	8503	74
2138	8504	65
2139	8505	80
213A	8506	69
213B	8507	77
213C	8508	66
213D	8509	72
213E	850A	6A
213F	850B	00
2140	850C	00

Line No	Assembler Message
0	Program executed successfully

Memory Locations :

8500H – 8509H : Temperature readings (10)

850AH : Threshold value

850BH : Number of abnormal readings

850CH : Highest temperature

3. An educational institution stores marks of ten students in memory. A student is considered successful if the mark is greater than or equal to the pass mark stored in memory. Develop an 8085 ALP to

- Count the number of students who passed.
- Find the highest mark.
- Calculate the total marks obtained by all students.
- Store all results in separate memory locations.

Load me at: 2000
Data Stack KeyPad Memory

```

1  LXI H, 8500H
2  MVI B, 0AH
3  MVI D, 00H
4  MVI E, 00H
5  MVI C, 00H
6  MVI A, 00H
7  STA 850BH
8
8  LOOP: MOV A, M
9        MOV L, A
10       ADD E
11       MOV E, A
12       JNC NOCARRY
13       INR D
14
15  NOCARRY:
16       MOV A, L
17       CMP C
18       JC CHECKPASS
19       MOV C, A
20
21  CHECKPASS:
22       MOV A, L
23       LDA 850AH
24       CMP A
25       JC NEXT
26       LDA 850BH
27       INR A
28       STA 850BH
29
30  NEXT:
31       INX H
32       DCR B
33       JNZ LOOP
34
35       MOV A, E
36       STA 850DH
37       MOV A, D
38       STA 850EH
39       MOV A, C
40       STA 850CH
41       HLT

```

Start 8500

Address (Hex)	Address	Data
2134	8500	45
2135	8501	68
2136	8502	72
2137	8503	59
2138	8504	83
2139	8505	75
213A	8506	64
213B	8507	91
213C	8508	52
213D	8509	77
213E	850A	50
213F	850B	00
2140	850C	00
2141	850D	00
2142	850E	00

Line No	Assembler Message
0	Program assembled successfully

Memory Locations :

8500H – 8509H : Marks of 10 students

850AH : Pass mark

850BH : Number of students passed

850CH : Highest mark

850DH – 850EH : Total marks (Low, High)